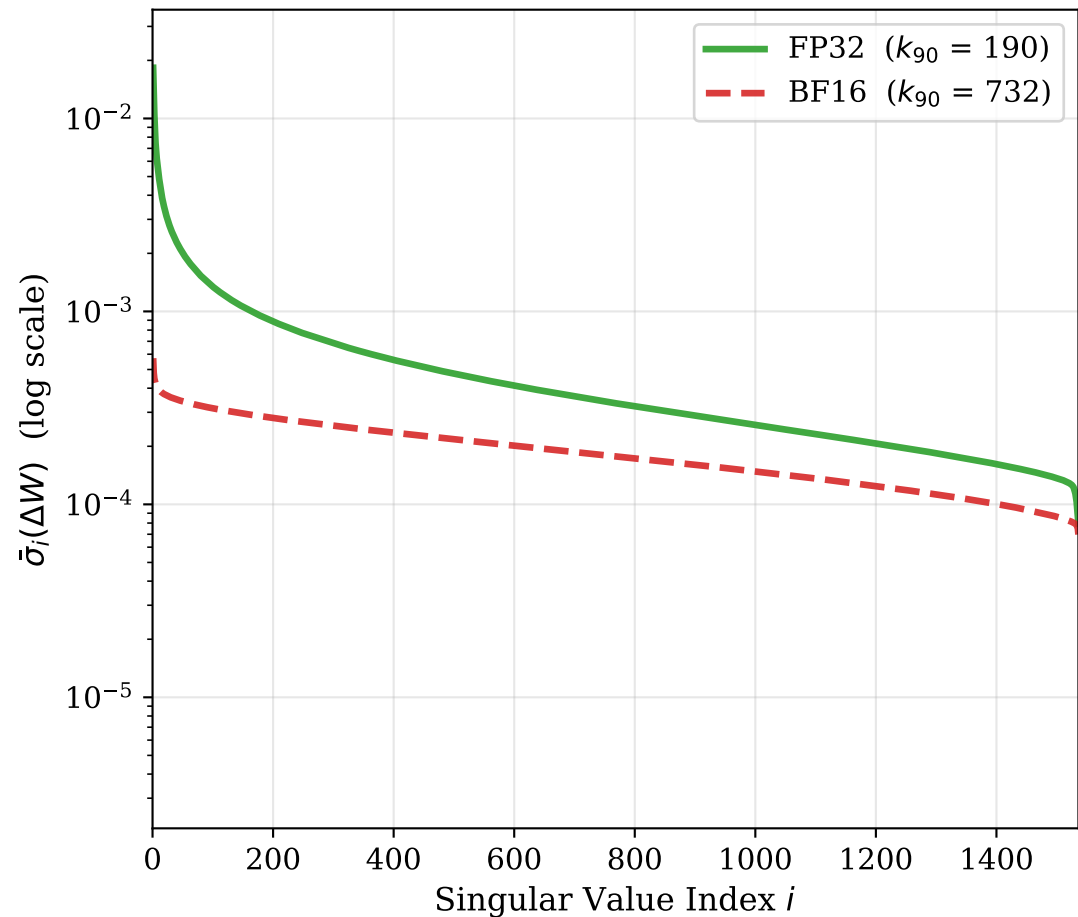


FP32 vs BF16 Spectral Decay of ΔW : How Precision Affects Spectral Structure

($\bar{\sigma}_i = \text{mean of } \sigma_i(\Delta W) \text{ across all layers and components}$)

AdamW + GRPO

FP32: concentrated (low-rank) | BF16: dispersed (high-rank)



Muon + GRPO

FP32: distributed | BF16: lower magnitude

